KTH AI SOCIETY · PYRMIT · PROJECT STATUS

What we built, what it needs to run, and what is left

A chat assistant for Swedish land use and building permits that answers from each kommun's own plans and from the law, and cites every claim.

THE IDEA

Ask about a place, get an answer from its plans and the law

Can I add a 25 m² guest house on my plot? (the place is pinned on the map)

Verdict: probably yes, without bygglov, if the plan's building-area limit for the plot still allows it [1]. A komplementbostadshus up to 30 m² inside a detaljplan needs no bygglov [2], but closer than 4.5 m to the boundary the neighbours must agree [3]…

[1] Detaljplan of the plot: plankarta and bestämmelser · [2] PBL 9 kap. 4 § · [3] PBL 9 kap. 34 §

Illustration of the answer format; each [n] opens the cited passage in its original PDF or statute.

What makes it hard

- **290 kommuner**, each with its own website, plans and PDFs, many of them scanned decades ago.
- The rules sit in **plan maps**: codes printed on zones of a drawing.
- **The law changes**: PBL 9 kap. was renumbered in 2025 (SFS 2025:974); the models remember the old version.
- Answers must be **sourced**: an unsupported claim is worse than no answer.

What it does

- Gives a verdict, explains a rule, proposes projects or develops an idea.
- Asks back when the place or the project is unclear.

WHAT WE BUILT

Four building blocks

Every number in this deck comes from a result file in the repository.

1 · Data

113k documents

from all 290 kommun websites and 15 statutes; 2.39M searchable passages

2 · Pipeline

11.4k scans read

crawl, OCR, filter, index, plan registry and checks, in one script on a GPU PC

3 · Agent

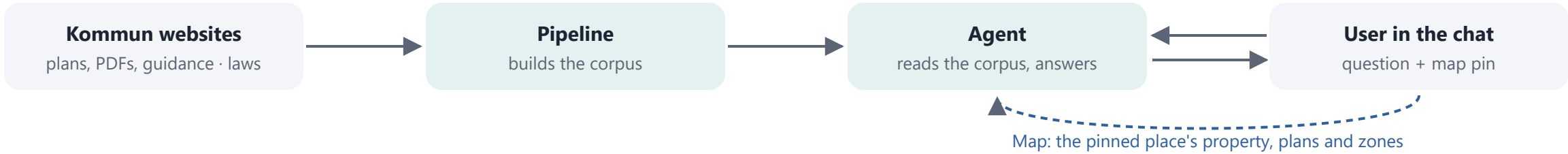
12 tools

searches, reads and looks at plan maps before it answers, citing each source; ~40 s, ~$0.01 an answer

4 · Map

2× better answers

a pinned place gives the agent the property, its plans and protected areas (Uppsala pilot)



1 · DATA

All of Sweden's planning documents, searchable

290
kommuner (all of Sweden), crawled politely: 1 request/s, robots.txt

89.5k
PDFs downloaded, 244 GB; 77.8k kept in the corpus

35k
kommun web pages (guidance, permits, fees)

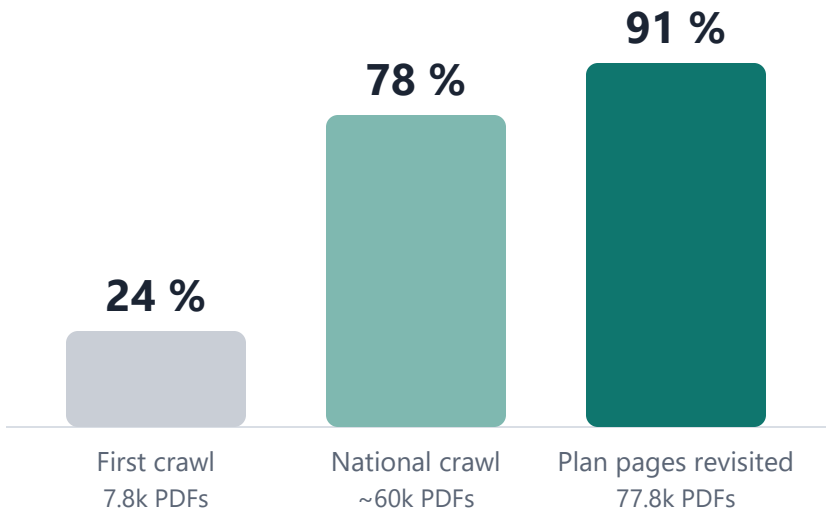
15
statutes from Riksdagen (PBL, PBF, MB, JB...), one passage per §

2.39M
passages embedded for search (exact duplicates removed)

38.7k
plans in the registry (279 kommuner); 97k coded rules read from 11.2k of them

Are the plans in it? Coverage benchmark

Share of the plans a kommun lists as in force that the corpus has (202 plans, 13 kommuner, fetched live)



Still missing: plans only shown in a **map application** (11 of the 29 sampled kommuner) and pages too deep for the crawl.

2 · PIPELINE

From websites to a checked corpus

Crawl
290 sites
revisit ~3 h

→

OCR
scanned PDFs
GPU, by priority

→

Extract
text per document
~25 min

→

Filter
drop off-topic PDFs
rules + model

→

Chunk + embed
passages → vectors
GPU, new ones only

→

Store + plans
search index,
plan registry

→

Checks
data, tool replay,
coverage

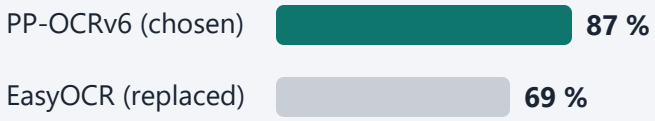
One script runs it: OCR until a set hour, then the rebuild (~3 h for 505k new passages), then the checks; a failed check keeps the previous corpus.

OCR: old plans become readable

40.5k pages

11.4k scanned PDFs in 222 kommuner, 10M words. Plan maps and plan texts are done; ~18 h of GPU left for 3.7k lower-priority documents.

Chosen on a benchmark



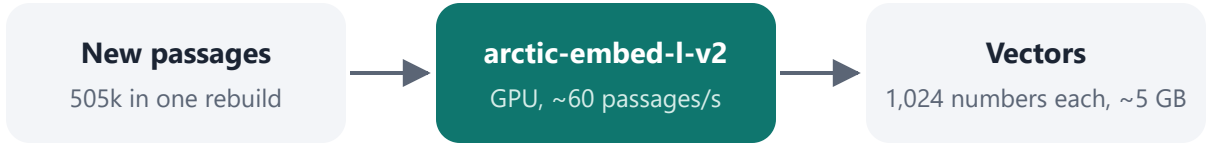
words read on hand-transcribed scans of plan maps

Checked, not trusted

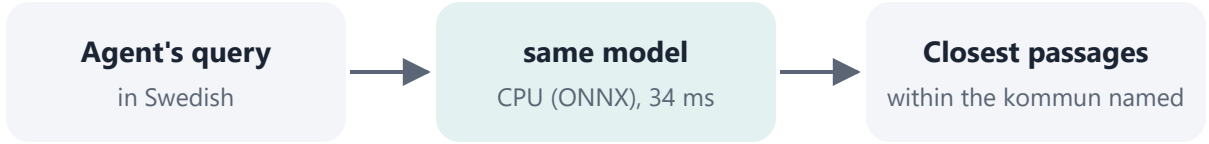
- Data checks after every rebuild
- 1,500 stored agent tool calls replayed
- Coverage against the kommuner's own plan lists
- A PDF is dropped only on a near-certain rule or by a second judge

Search runs on an open embedding model we host ourselves

OFFLINE · AT EACH REBUILD · GPU



ONLINE · EACH SEARCH · CPU



0.92
recall@5 on 45 Swedish questions: as good as the best API models, best at rank 1

0.9 GB
RAM on a CPU, same recall as the GPU: the server needs no GPU

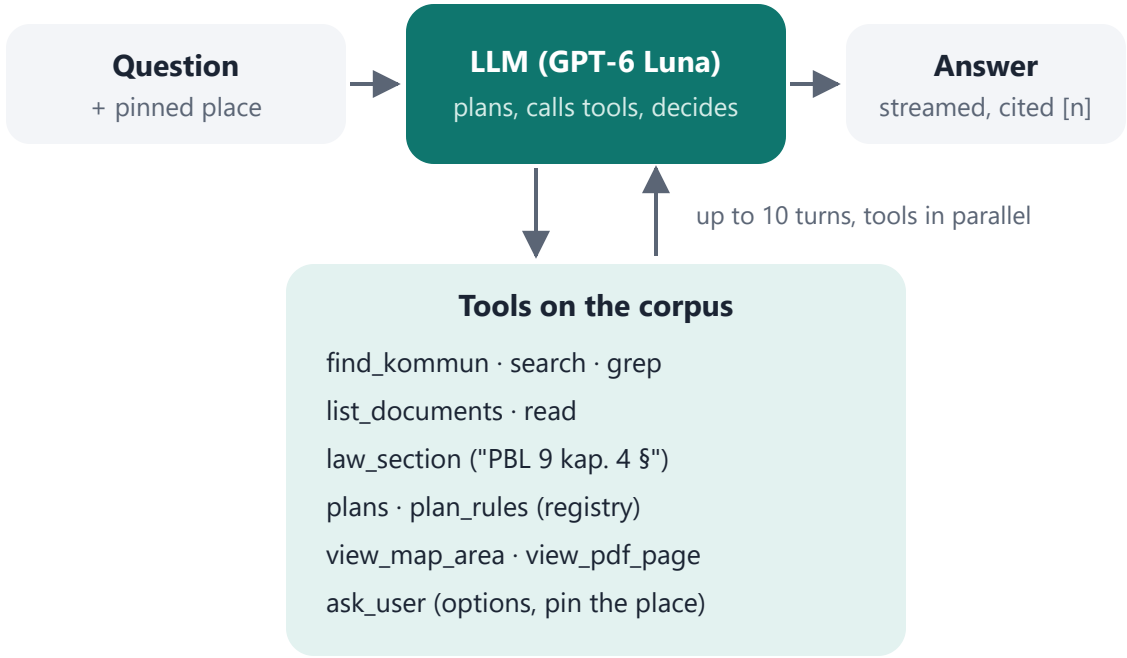
$0
open weights (Apache-2.0), 568M parameters, multilingual

Or through the LLM gateway?

Corpus and queries must use the same model: switching means embedding the whole corpus again, once.

	LOCAL (IN USE)	VIA A GATEWAY
Model	arctic-embed-l-v2 (not on OpenRouter)	bge-m3, multilingual-e5-large or Qwen3-8B on OpenRouter: recall 0.91–0.92, within noise
Cost	free; GPU time	~$7 to embed the corpus once (~685M tokens at $0.01/M), ~$1 per rebuild after
Hosting	model in the server's RAM; bulk indexing needs a GPU	nothing to host; indexing needs no GPU (only the OCR does)
Risk	none new	recall to measure again on the full corpus (~$2.5); Neon's gateway only has Qwen3-0.6B (0.81) and English models

An agent that looks things up before it answers

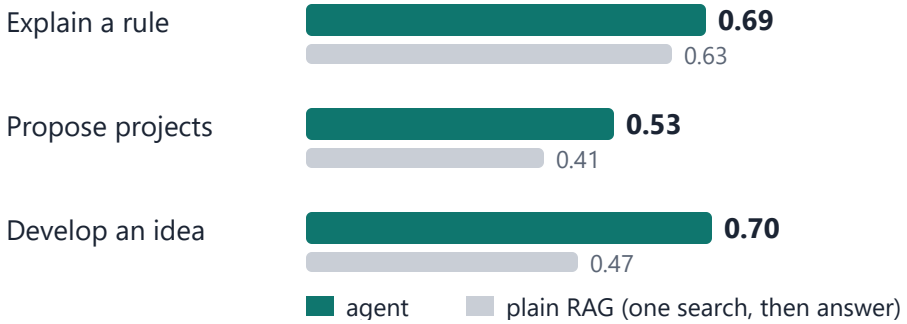


Each [n] opens the passage highlighted in its original PDF;
the "Inspect pipeline" panel shows every model turn and tool call.

Measured on ~370 benchmark cases

6 sets: feasibility, real MÖD rulings, plan maps, explain / propose / develop tasks, generated cases, pinned places. Two judges from different model families (agreement κ = 0.89).

Share of the reference key points the answer covers



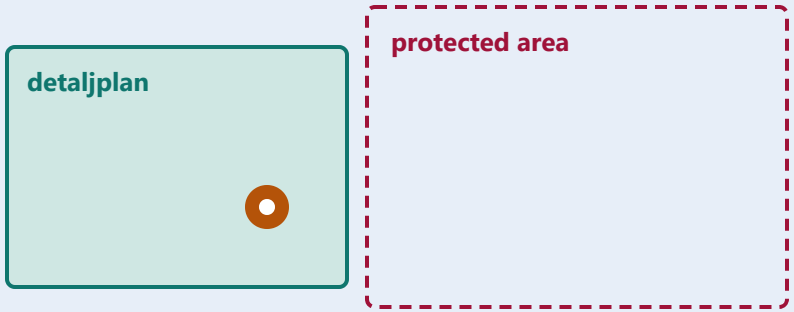
~40 s
per answer, first words ~25 s

~$0.01
per answer (LLM tokens)

1.45
unsupported claims per answer: the next target

4 · MAP

Pin a place: the agent knows the property and its rules



Place card

Property and its addresses
Plans, newest first; inside or outside a plan
Protected areas, Natura 2000, national interests
Ancient remains (live lookup)
Permits likely needed (fixed rules, no LLM)

0.79–0.82
key points covered with a pin, vs **0.39** with only the address in the question (15 Uppsala places × 3)

0.3 s
to build a place card (median), from map-layer snapshots plus live lookups

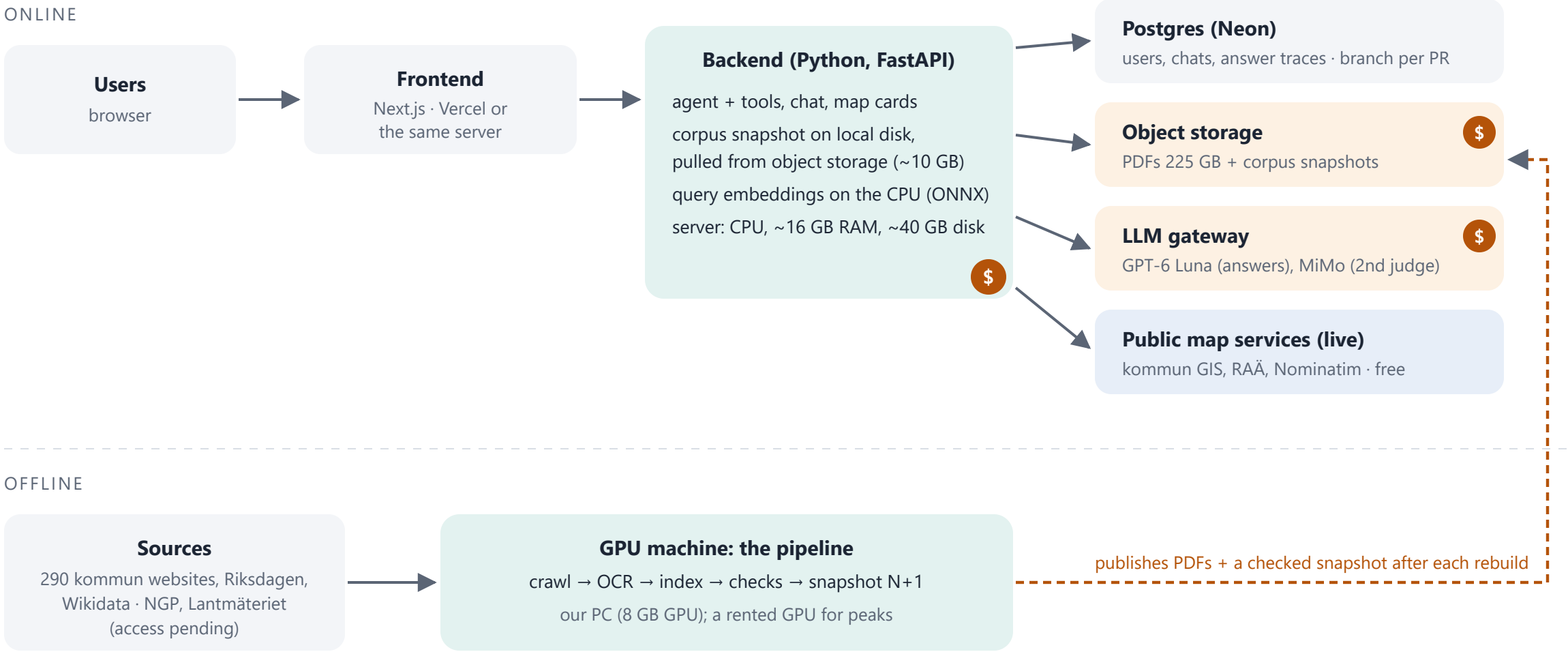
328 / 330
points where the snapshots agree with the providers' own queries

100 %
of typed addresses land on the right property, using the kommun's address register

Pilot: **Uppsala**, from the kommun's map service: 1,410 plans and their 3,930 PDFs now also in the corpus. Next kommuner need **NGP** (national digital plans) and **Lantmäteriet** (properties): access requested, not granted yet.

WHAT IT NEEDS TO RUN

Five services online, one GPU machine offline



HOSTING OPTIONS

Three ways to provide those services

	A · BEST OF EACH (RECOMMENDED)	B · EVERYTHING ON NEON	C · MANAGED CLOUD
LLM	OpenRouter (in use)	Neon AI Gateway	Vercel AI Gateway (already supported)
Files	Cloudflare R2	Neon Object Storage	AWS S3
Database	Neon Postgres (in use)	Neon Postgres	Neon Postgres
Embeddings	local on the server (CPU), or OpenRouter	local: Neon's embedding models are weaker for Swedish	local, or the gateway
Backend	one server (e.g. Hetzner, EU)	one server: Neon Functions run Node, our backend is Python	managed containers (Fly, Railway, Cloud Run)
Per month	**~€20–35** + LLM (+ OpenRouter's fee on credits)	**~€22–37** + LLM at list price	**≥ $45** (16 GB container, to price) + LLM
For	every model, incl. MiMo (2nd judge, PDF filter); a capped key per person; cost of each call; mature	one bill; no markup on models; files branch with the database (a PR branch sees its own corpus)	no server to look after; scales up
Against	four accounts; a server to look after	just out of beta; no MiMo; default caps $20/day and 200k tokens/min; no cost per call	dearest; S3 charges egress ($0.09/GB); more to configure

Files for 225 GB: R2 ~$3.2/month (no egress fee), Neon ~$5.2/month (500 GB egress included). The LLM dominates either way: ~$10 per 1,000 answers. Server prices to check before ordering.

HOW WE WORK

One path for code, one for data, both through staging

CODE · people and coding agents alike

Linear issue — every task, also agents'
→ **Branch + worktree** — own Neon branch, own ports
→ **Pull request** — CI: lint + backend tests
→ **Human review** — also for agents' PRs
→ **Staging**
→ **Production**

DATA · on the GPU machine

Crawl — revisit of known pages
→ **OCR + rebuild** — one script
→ **Free checks** — data, replay, coverage
→ **Snapshot N+1** — to object storage
→ **Staging** — paid benchmarks
→ **Production**

a failed check stops the release, not the pipeline

Capped LLM keys
one per person, agent and environment: a runaway benchmark only spends its own cap

Results in git
every paid run has a label; its JSON is committed and the reports are generated from it

Shared corpus
built once, downloaded by everyone; a bug report names its snapshot

WHAT IS LEFT

Remaining work, by stream

STREAM	TASK	WHY	EFFORT · COST
Client	**Talk to the client**: who the users are, their real questions, which kommuner first; have them check a sample of our benchmark cases	we have built without user feedback; the cases are written by us and by models	a meeting · free
Data	**Finish the OCR** (3.7k lower-priority scans)	the rest of the scans become searchable	~18 h GPU · free
	Full re-crawl; plans of map-only kommuner from their map services, as for Uppsala	66 of the 79 plan PDFs still missing sit on pages too deep; 11 of 29 kommuner list plans only in a map	several hours + per kommun · free
	NGP and Lantmäteriet once access is granted	structured plan rules, official plan IDs, a complete list to measure coverage	after approval · to check
Agent	**Check each claim** against its cited passage before the answer is sent	1.45 unsupported claims per answer	days · ~$2 per test round
	Better benchmarks: expert-reviewed references, multi-turn cases, pinned places beyond Uppsala, plan-map zones	the scores are only as good as the references	ongoing · ~$2–3 per round
Platform	**Host it** (option A or B), publish corpus snapshots, backend tests in CI, the pipeline script versioned	the app and the pipeline run on one PC	~1 week · ~€20–35/month